

Data-Exploration.R

forrest

2026-02-01

```
# Data Loading + Cleaning
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.6
```

```
## v forcats    1.0.1      v stringr   1.6.0
```

```
## v ggplot2    4.0.1      v tibble    3.3.1
```

```
## v lubridate  1.9.4      v tidyr     1.3.2
```

```
## v purrr      1.2.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(janitor)
```

```
##
```

```
## Attaching package: 'janitor'
```

```
##
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      chisq.test, fisher.test
```

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
file_path <- "group_stage_qualification_data.csv"
```

```
raw <- readr::read_csv(file_path, col_names = FALSE, show_col_types = FALSE)
```

```
hdr <- raw %>% slice(2) %>% unlist(use.names = FALSE) %>% as.character()
```

```
dat <- raw %>%  
  slice(-(1:2)) %>%  
  as_tibble()
```

```
names(dat) <- make.names(hdr, unique = TRUE)
```

```
dat <- dat %>% filter(dat[[6]] != "qualified_from_group")
```

```
stopifnot(ncol(dat) == 21)
```

```
names(dat) <- c(  
  "tournament_name",  
  "tournament_year",
```

```

"group_id",
"team_name",
"confederation",
"qualified_from_group",
"tournament_start_date",
"team_elo_pre",
"fifa_rank_pre",
"recent_window_n_matches",
"recent_win_rate",
"recent_goal_diff_per_match",
"recent_goals_for_per_match",
"recent_goals_against_per_match",
"opp_elo_mean",
"opp_elo_max",
"group_elo_std",
"elo_gap_vs_opp_mean",
"host_team_flag",
"gdp_per_capita_pre",
"log_gdp_per_capita_pre"
)

dat <- dat %>%
  mutate(
    tournament_year = as.integer(tournament_year),
    qualified_from_group = as.integer(qualified_from_group),
    host_team_flag = as.integer(host_team_flag),
    tournament_start_date = as.Date(tournament_start_date),
    team_elo_pre = as.numeric(team_elo_pre),
    fifa_rank_pre = as.numeric(fifa_rank_pre),
    recent_window_n_matches = as.numeric(recent_window_n_matches),
    recent_win_rate = as.numeric(recent_win_rate),
    recent_goal_diff_per_match = as.numeric(recent_goal_diff_per_match),
    recent_goals_for_per_match = as.numeric(recent_goals_for_per_match),
    recent_goals_against_per_match = as.numeric(recent_goals_against_per_match),
    opp_elo_mean = as.numeric(opp_elo_mean),
    opp_elo_max = as.numeric(opp_elo_max),
    group_elo_std = as.numeric(group_elo_std),
    elo_gap_vs_opp_mean = as.numeric(elo_gap_vs_opp_mean),
    gdp_per_capita_pre = as.numeric(gdp_per_capita_pre),
    log_gdp_per_capita_pre = as.numeric(log_gdp_per_capita_pre)
  )

cat("Rows:", nrow(dat), "Cols:", ncol(dat), "\n")

## Rows: 224 Cols: 21
cat("Class balance (qualified_from_group):\n")

## Class balance (qualified_from_group):
print(prop.table(table(dat$qualified_from_group)))

##
##    0    1
## 0.5 0.5

```

```

print(head(dat, 3))

## # A tibble: 3 x 21
##   tournament_name      tournament_year group_id team_name confederation
##   <chr>                <int> <chr>      <chr>      <chr>
## 1 1998 FIFA Men's World Cup      1998 Group A Brazil    CONMEBOL
## 2 1998 FIFA Men's World Cup      1998 Group A Morocco  CAF
## 3 1998 FIFA Men's World Cup      1998 Group A Norway    UEFA
## # i 16 more variables: qualified_from_group <int>,
## #   tournament_start_date <date>, team_elo_pre <dbl>, fifa_rank_pre <dbl>,
## #   recent_window_n_matches <dbl>, recent_win_rate <dbl>,
## #   recent_goal_diff_per_match <dbl>, recent_goals_for_per_match <dbl>,
## #   recent_goals_against_per_match <dbl>, opp_elo_mean <dbl>,
## #   opp_elo_max <dbl>, group_elo_std <dbl>, elo_gap_vs_opp_mean <dbl>,
## #   host_team_flag <int>, gdp_per_capita_pre <dbl>, ...

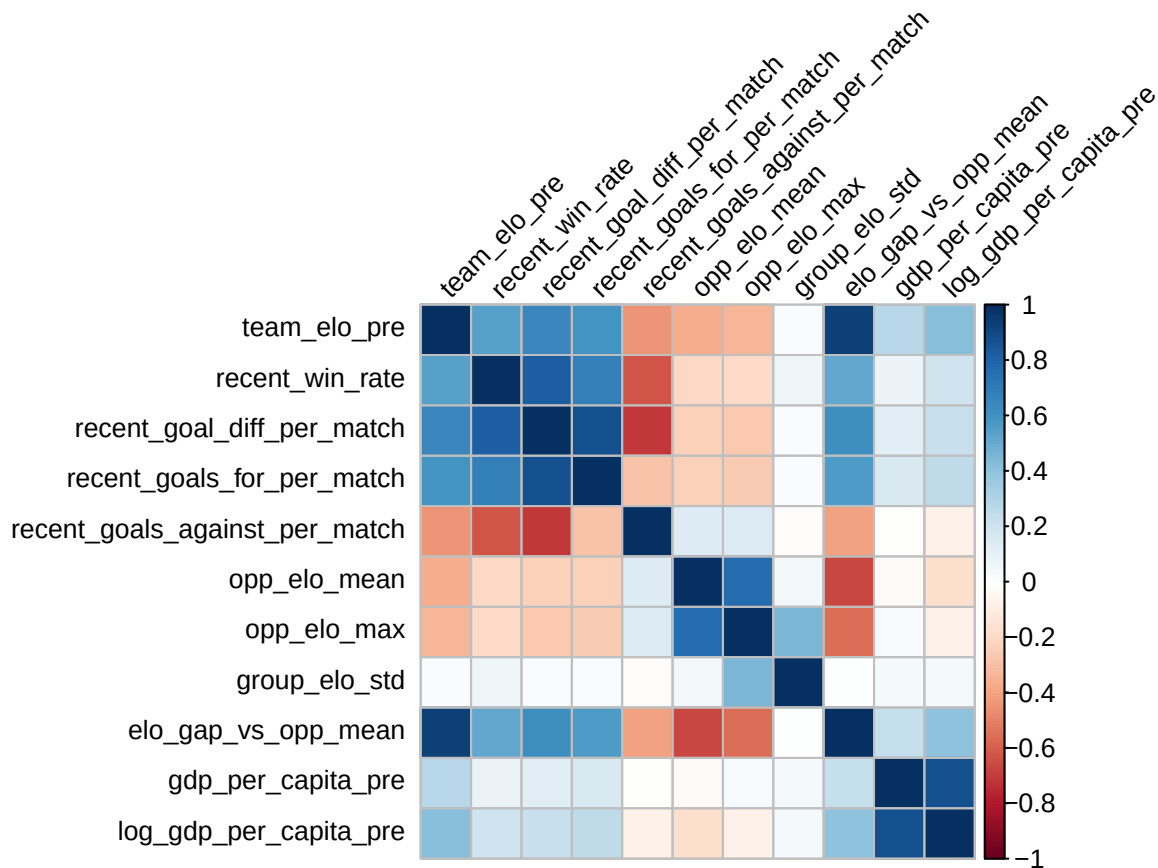
# Correlations among numeric predictors

num_cols <- c(
  "team_elo_pre", "fifa_rank_pre", "recent_window_n_matches", "recent_win_rate",
  "recent_goal_diff_per_match", "recent_goals_for_per_match", "recent_goals_against_per_match",
  "opp_elo_mean", "opp_elo_max", "group_elo_std", "elo_gap_vs_opp_mean",
  "gdp_per_capita_pre", "log_gdp_per_capita_pre"
)

num_df <- dat %>%
  select(all_of(num_cols)) %>%
  select(where(~ sum(!is.na(.x)) > 5)) %>%
  select(where(~ sd(.x, na.rm = TRUE) > 0))

cor_mat <- cor(num_df, use = "pairwise.complete.obs")
corrplot(
  cor_mat,
  method = "color",
  type = "full",
  tl.cex = 0.8,
  tl.col = "black",
  tl.srt = 45,
  addgrid.col = "gray75"
)

```



```
cor_long <- as.data.frame(as.table(cor_mat)) %>%
  rename(var1 = Var1, var2 = Var2, r = Freq) %>%
  mutate(
    var1 = as.character(var1),
    var2 = as.character(var2),
    abs_r = abs(r)
  ) %>%
  filter(var1 != var2) %>%      # remove self-correlations
  distinct(pmin(var1, var2), pmax(var1, var2), .keep_all = TRUE) %>%
  arrange(desc(abs_r))
```

```
cat("\nTop 15 strongest predictor-predictor correlations:\n")
```

```
##
## Top 15 strongest predictor-predictor correlations:
```

```
print(head(cor_long, 15))
```

```
##
##          var1                                var2      r
## 1      elo_gap_vs_opp_mean                    team_elo_pre 0.9370858
## 2  recent_goals_for_per_match recent_goal_diff_per_match 0.8794918
## 3      log_gdp_per_capita_pre                    gdp_per_capita_pre 0.8721555
## 4  recent_goal_diff_per_match                    recent_win_rate 0.8222426
## 5              opp_elo_max                      opp_elo_mean 0.7687910
## 6 recent_goals_against_per_match recent_goal_diff_per_match -0.7054836
## 7      recent_goals_for_per_match                    recent_win_rate 0.6898679
```

```

## 8          elo_gap_vs_opp_mean          opp_elo_mean -0.6668067
## 9      recent_goal_diff_per_match          team_elo_pre  0.6592778
## 10 recent_goals_against_per_match          recent_win_rate -0.6296506
## 11          elo_gap_vs_opp_mean recent_goal_diff_per_match  0.6175149
## 12      recent_goals_for_per_match          team_elo_pre  0.5965697
## 13          elo_gap_vs_opp_mean recent_goals_for_per_match  0.5637284
## 14          elo_gap_vs_opp_mean          opp_elo_max -0.5527474
## 15          recent_win_rate          team_elo_pre  0.5480606
##      abs_r          pmin(var1, var2)          pmax(var1, var2)
## 1  0.9370858          elo_gap_vs_opp_mean          team_elo_pre
## 2  0.8794918      recent_goal_diff_per_match      recent_goals_for_per_match
## 3  0.8721555          gdp_per_capita_pre          log_gdp_per_capita_pre
## 4  0.8222426      recent_goal_diff_per_match          recent_win_rate
## 5  0.7687910          opp_elo_max          opp_elo_mean
## 6  0.7054836      recent_goal_diff_per_match recent_goals_against_per_match
## 7  0.6898679      recent_goals_for_per_match          recent_win_rate
## 8  0.6668067          elo_gap_vs_opp_mean          opp_elo_mean
## 9  0.6592778      recent_goal_diff_per_match          team_elo_pre
## 10 0.6296506 recent_goals_against_per_match          recent_win_rate
## 11 0.6175149          elo_gap_vs_opp_mean      recent_goal_diff_per_match
## 12 0.5965697      recent_goals_for_per_match          team_elo_pre
## 13 0.5637284          elo_gap_vs_opp_mean      recent_goals_for_per_match
## 14 0.5527474          elo_gap_vs_opp_mean          opp_elo_max
## 15 0.5480606          recent_win_rate          team_elo_pre

# Correlations with qualification outcome

safe_cor <- function(x, y) {
  x <- suppressWarnings(as.numeric(x))
  y <- suppressWarnings(as.numeric(y))
  ok <- is.finite(x) & is.finite(y)
  if (sum(ok) < 5) return(NA_real_)
  if (sd(x[ok]) == 0) return(NA_real_)
  cor(x[ok], y[ok])
}

outcome_cor <- tibble(
  feature = num_cols,
  r = purrr::map_dbl(num_cols, ~ safe_cor(dat[[.x]], dat$qualified_from_group))
) %>%
  mutate(abs_r = abs(r)) %>%
  arrange(desc(abs_r))

cat("\nCorrelations with qualified_from_group (sorted):\n")

##
## Correlations with qualified_from_group (sorted):
print(outcome_cor)

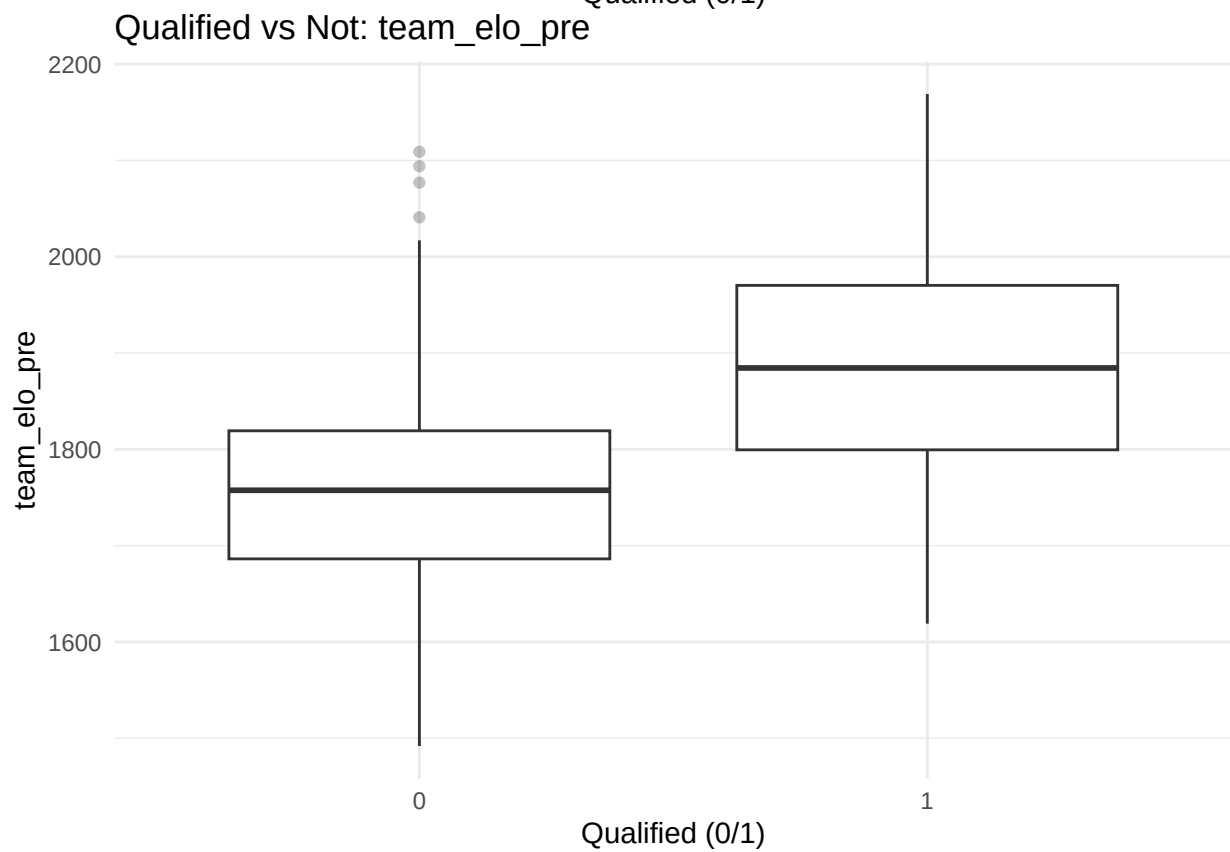
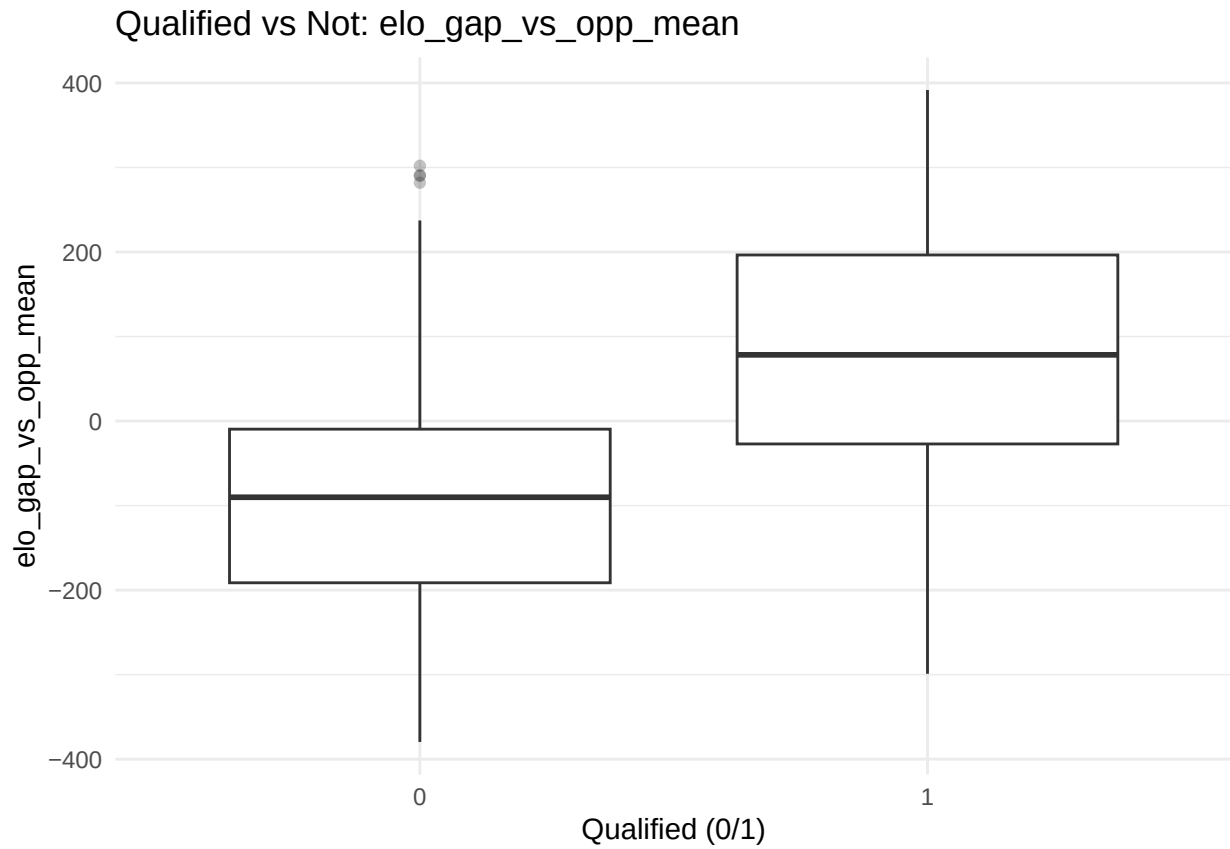
## # A tibble: 13 x 3
##   feature          r abs_r
##   <chr>          <dbl> <dbl>
## 1 elo_gap_vs_opp_mean  0.479  0.479
## 2 team_elo_pre        0.449  0.449

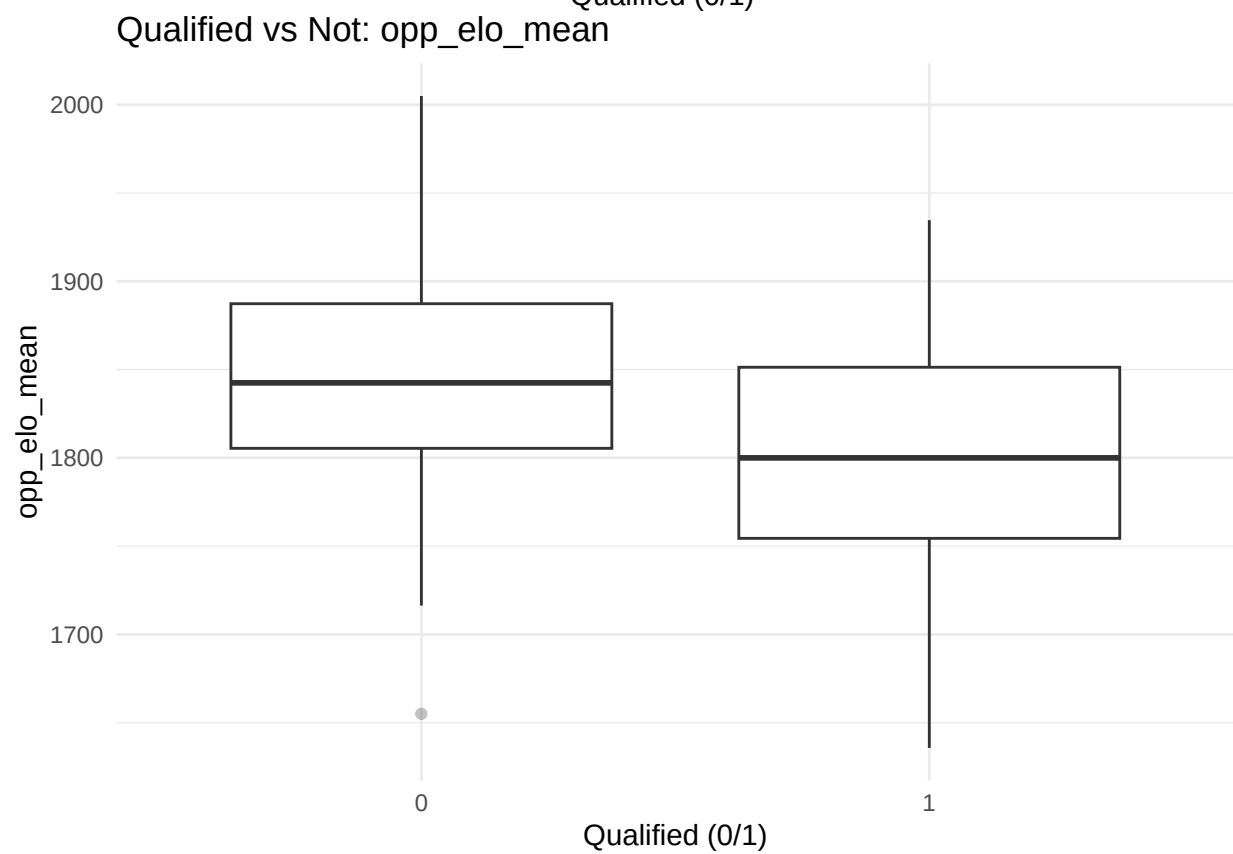
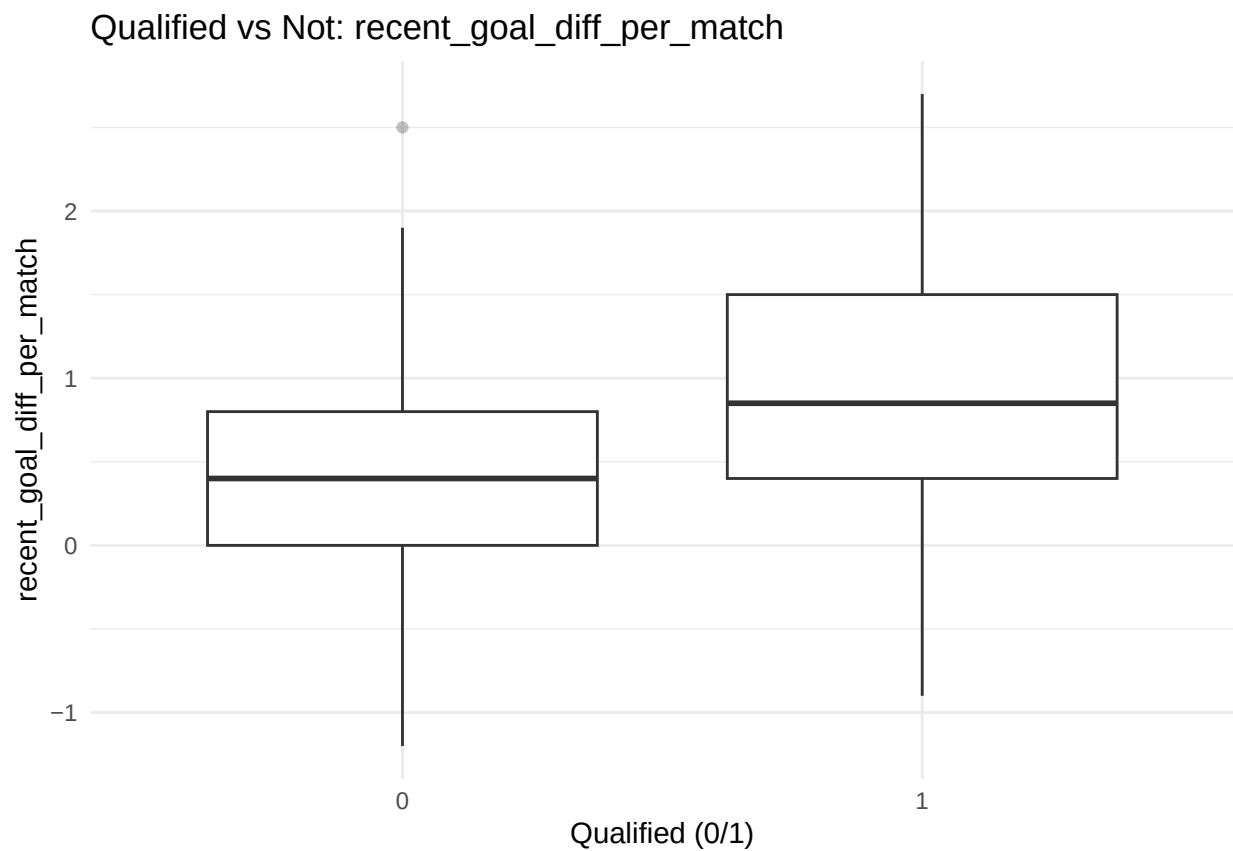
```

```
## 3 recent_goal_diff_per_match      0.331  0.331
## 4 opp_elo_mean                    -0.319  0.319
## 5 recent_goals_for_per_match      0.314  0.314
## 6 recent_win_rate                 0.278  0.278
## 7 log_gdp_per_capita_pre          0.255  0.255
## 8 opp_elo_max                     -0.225  0.225
## 9 recent_goals_against_per_match -0.200  0.200
## 10 gdp_per_capita_pre             0.194  0.194
## 11 group_elo_std                   0      0
## 12 fifa_rank_pre                  NA     NA
## 13 recent_window_n_matches        NA     NA
```

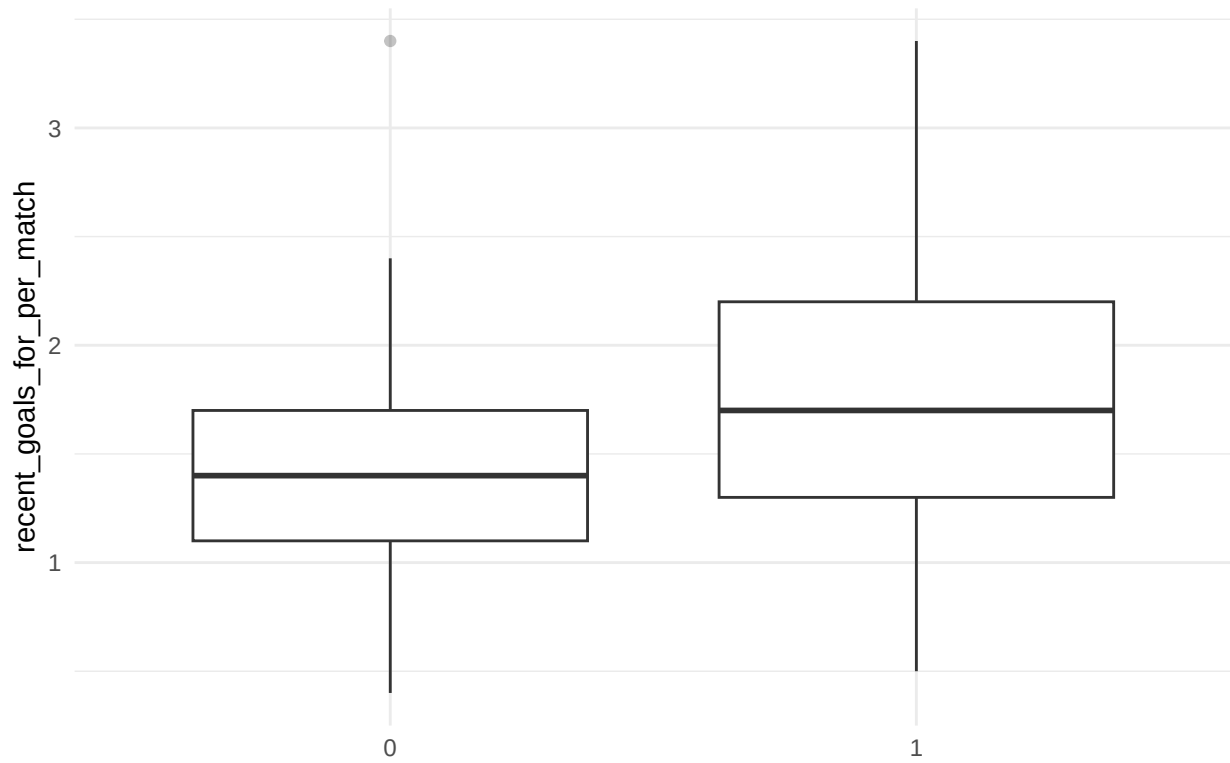
```
top_feats <- outcome_cor %>%
  filter(!is.na(r)) %>%
  slice_head(n = 6) %>%
  pull(feature)
```

```
for (f in top_feats) {
  p <- ggplot(dat, aes(x = factor(qualified_from_group), y = suppressWarnings(as.numeric(.data[[f]]))))
  geom_boxplot(outlier.alpha = 0.3) +
  theme_minimal() +
  labs(
    title = paste("Qualified vs Not:", f),
    x = "Qualified (0/1)",
    y = f
  )
  print(p)
}
```

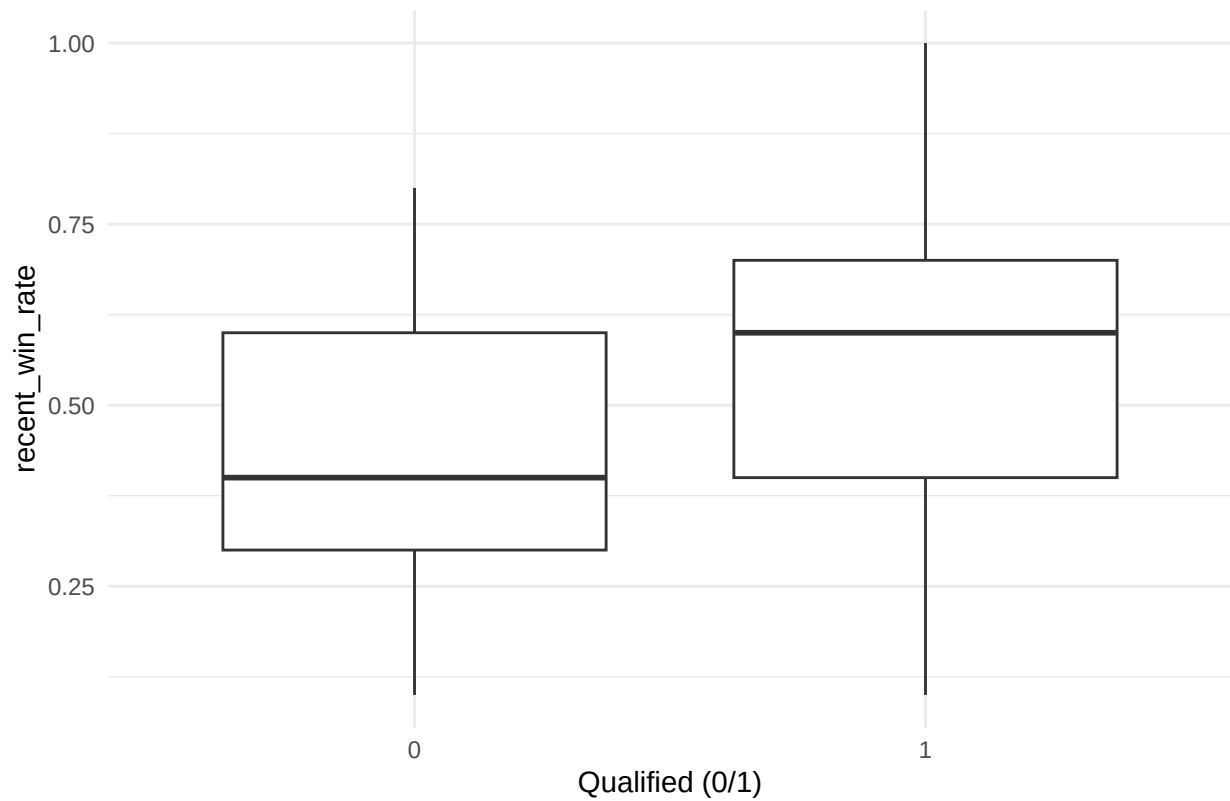




Qualified vs Not: recent_goals_for_per_match



Qualified vs Not: recent_win_rate



```

# Data Exploration

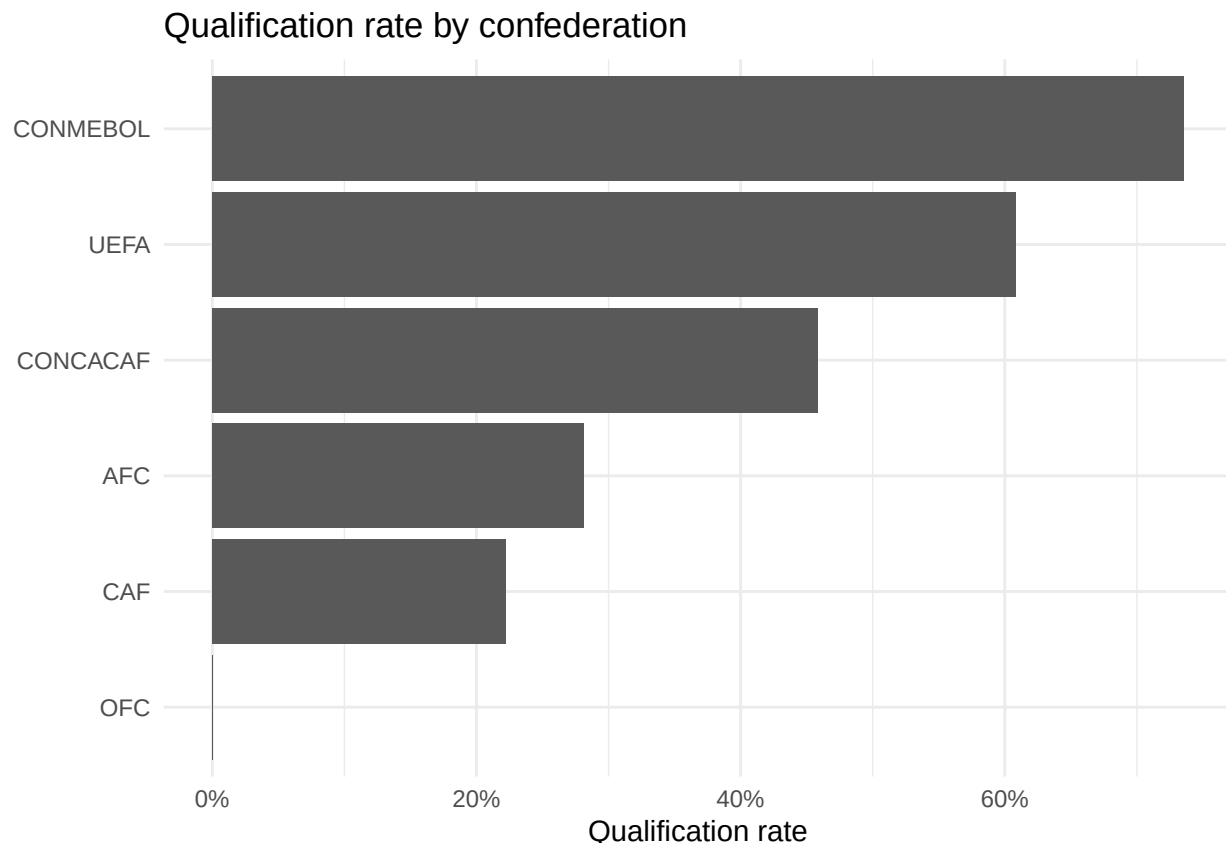
conf_rates <- dat %>%
  group_by(confederation) %>%
  summarise(
    n = n(),
    qualify_rate = mean(qualified_from_group, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(desc(qualify_rate))

print(conf_rates)

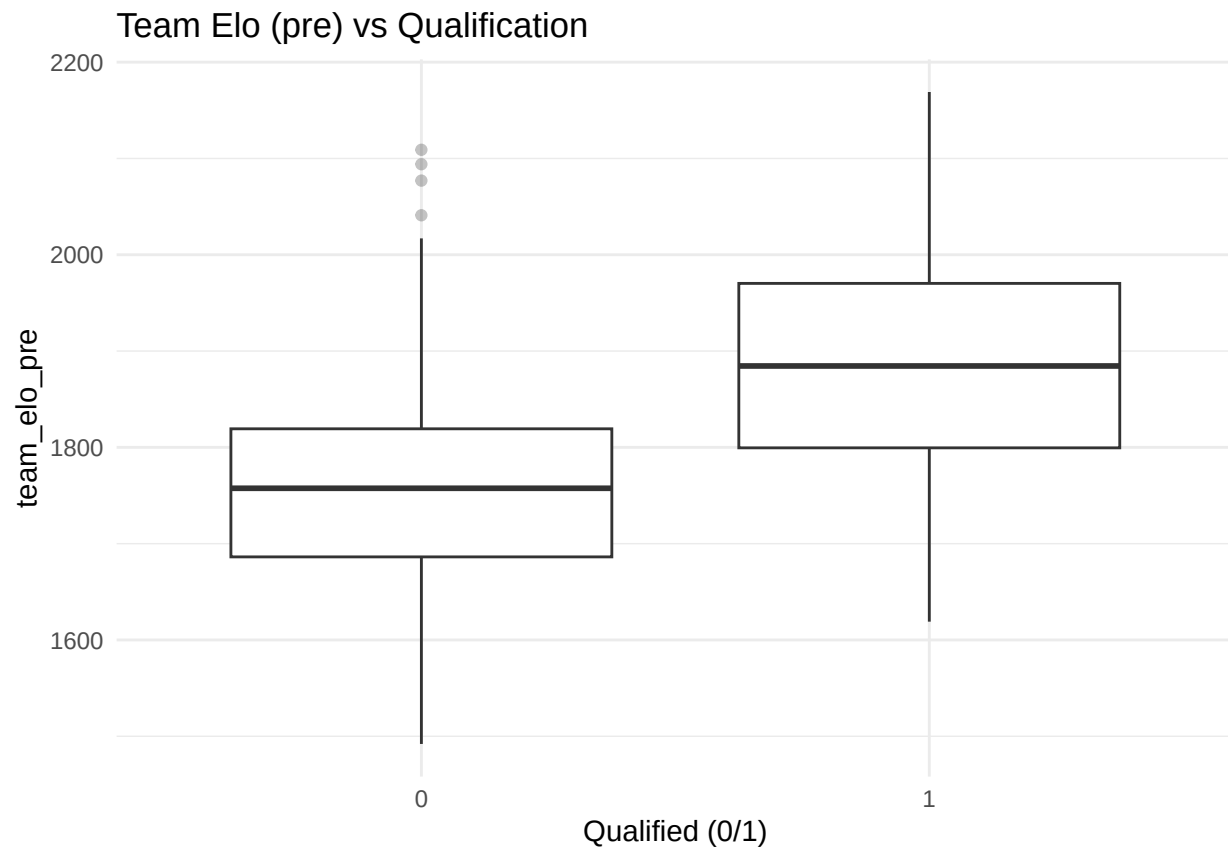
## # A tibble: 6 x 3
##   confederation      n qualify_rate
##   <chr>          <int>         <dbl>
## 1 CONMEBOL        34          0.735
## 2 UEFA           97          0.608
## 3 CONCACAF        24          0.458
## 4 AFC            32          0.281
## 5 CAF            36          0.222
## 6 OFC             1           0

p_conf <- ggplot(conf_rates, aes(x = reorder(confederation, qualify_rate), y = qualify_rate)) +
  geom_col() +
  coord_flip() +
  scale_y_continuous(labels = scales::percent_format()) +
  theme_minimal() +
  labs(title = "Qualification rate by confederation", x = NULL, y = "Qualification rate")
print(p_conf)

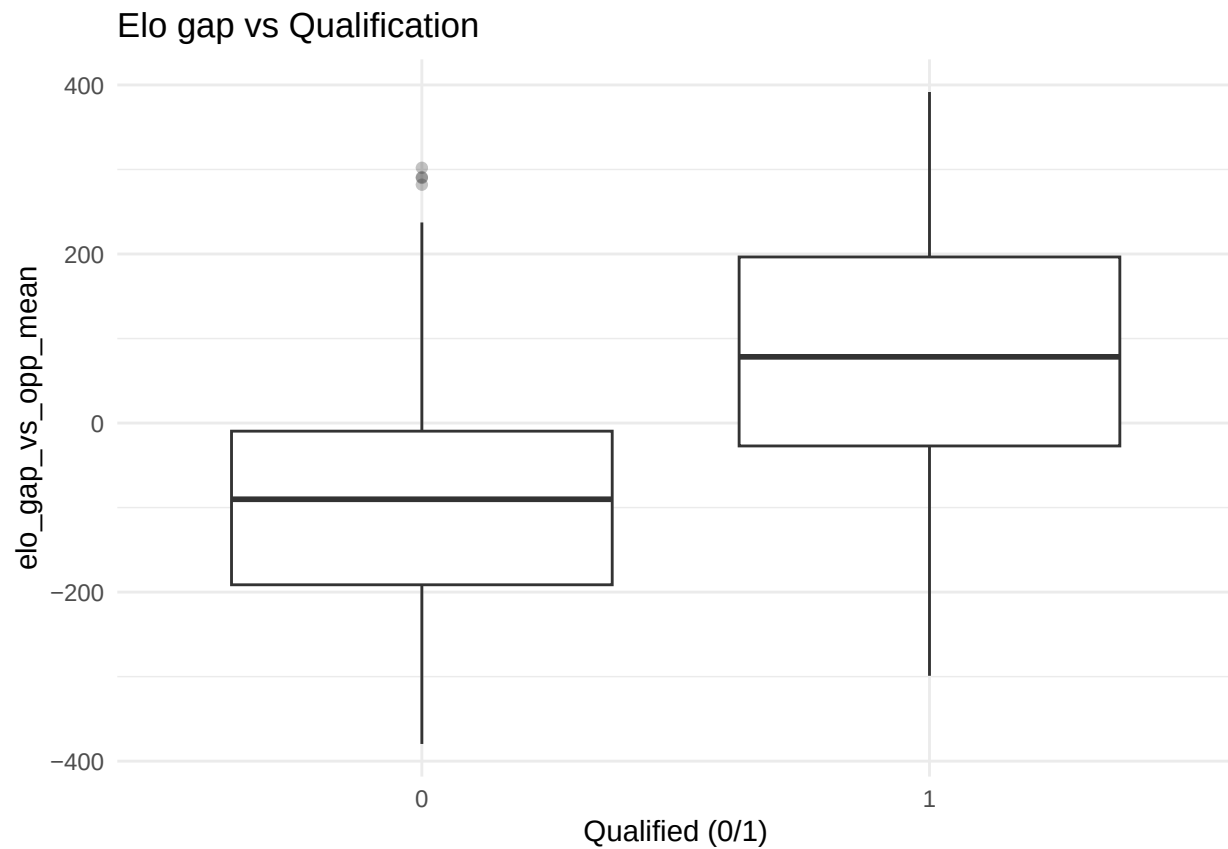
```



```
p_elo <- ggplot(dat, aes(x = factor(qualified_from_group), y = team_elo_pre)) +  
  geom_boxplot(outlier.alpha = 0.3) +  
  theme_minimal() +  
  labs(title = "Team Elo (pre) vs Qualification", x = "Qualified (0/1)", y = "team_elo_pre")  
print(p_elo)
```



```
p_gap <- ggplot(dat, aes(x = factor(qualified_from_group), y = elo_gap_vs_opp_mean)) +  
  geom_boxplot(outlier.alpha = 0.3) +  
  theme_minimal() +  
  labs(title = "Elo gap vs Qualification", x = "Qualified (0/1)", y = "elo_gap_vs_opp_mean")  
print(p_gap)
```



```
p_opp <- ggplot(dat, aes(x = factor(qualified_from_group), y = opp_elo_mean)) +  
  geom_boxplot(outlier.alpha = 0.3) +  
  theme_minimal() +  
  labs(title = "Opponent mean Elo vs Qualification", x = "Qualified (0/1)", y = "opp_elo_mean")  
print(p_opp)
```

